

The Policy-Class Lattice: Which Restriction Destroyed Viability?

Edition 2, revised after a joint audit

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September 24, 2026

Abstract

Which restriction destroyed viability — the codex, the reading, the instrument set, or the belief? The programme’s papers separate policy classes case by case; this paper, the D7 item of the generalization catalogue, organizes the audited classes as a lattice and computes the diagnostic exactly. The two restriction axes — observation reading (aggregate, full) and action discipline (no-repeat codex, free) — generate four classes whose kernels on the two-patch pair family hold exactly 24, 26, 25, and 28 of the 36 safe pairs: a genuine product lattice with incomparable middle cells (the joint audit’s missing fourth cell, computed here), the institutional chain $24 \subseteq 26 \subseteq 28$ running along its lower-left edge, the kernel meet of the middle cells equal to the institutional kernel setwise, and the kernel join the full-information kernel. Viability kernels are monotone under class inclusion throughout, and the certainty-equivalence separation mechanism of the successor’s trap (the canonical uncorrected law’s drift $\geq \frac{21}{100}$, against the corrected law’s identically zero drift) is re-verified as the off-grid separator. The diagnostic contribution is *distance to viability*: for the audited nonviable belief, the minimum sensor refinement that restores viability is exactly one split, the minimum action-set expansion is exactly one instrument (which restores viability under the unchanged reading, enlarging the viable family to all 36 pairs), and the minimum prior shrinking is exactly one state. Every number is machine-verified in exact arithmetic (6/6 check families).

Introduction and relation to the programme

The completed items distinguish perfect-information policies, observation-based policies, certainty-equivalence policies, and institutionally restricted policies as separate mechanisms. The source catalogue proposes the unified reading adopted here: a parameterized class Π with kernels $\text{Viab}_{\Pi}(\mathcal{V})$, monotone under inclusion, forming a hierarchy — and, more consequentially for governance, a diagnostic question: which restriction is responsible for the loss? The present paper instantiates that programme on

the audited working system, computing the inclusions and the distances exactly. No completed item is modified; the stochastic selector’s probabilistic class and the successor’s five layers are cited where they bound the reading.

Policy classes and the chain

A policy class Π is a family of observation-based laws (measurable in the reading, acting per cell under the successor’s read-then-act semantics — the cell’s single action is chosen on the reading and applied to every compatible state). Its kernel is the set of beliefs from which some law of the class maintains the floors indefinitely, under the standing adversarial-disturbance reading of the calculus (Abaee 2026a, Section 2: maintenance against every admissible disturbance). The pair family: the $\binom{9}{2} = 36$ two-state beliefs of the nine floor-safe states (z_1, z_2) , $z_i \in \{1, 2, 3\}$, of the two-patch working system (Abaee 2026a, Section 8; full transition data in the script). Under full information a pair is viable exactly when both of its states are; the corner $(1, 1)$ is dynamically nonviable — no single instrument keeps both floors against damage 2 — whence the full-information kernel is exactly the $\binom{8}{2} = 28$ pairs avoiding $(1, 1)$. The *no-repeat codex* is the institutional action discipline requiring the per-cell action to differ from the previously executed action ($u_t \neq u_{t-1}$; the first step unconstrained).

Proposition 1 (Monotonicity). $\Pi_1 \subseteq \Pi_2 \implies \text{Viab}_{\Pi_1}(\mathcal{V}) \subseteq \text{Viab}_{\Pi_2}(\mathcal{V})$ (the policy-class instance of the epistemic kernel’s monotonicity, Abaee 2026a, Proposition 8; the selector’s Theorem 2(iii) and Corollary 1 carry the same statement — restated here as the lattice’s order lemma).

Proof. A witness law for B under Π_1 is a law of Π_2 . $\square$

Proposition 2 (The audited chain is strict). On the two-patch pair family: the institutional class (aggregate reading with a no-repeat codex) sustains 24 of 36 safe pairs; the per-cell aggregate class sustains 26; the full-information class sustains 28. Each inclusion is setwise and strict, with separating beliefs $\{(1, 2), (3, 1)\}$ at the lower link and $\{(1, 2), (2, 1)\}$ at the upper. On this family the certainty-equivalence class coincides with the per-cell class (laws act

on the reading only); the successor’s trap — the canonical uncorrected law’s drift $\geq \frac{21}{100}$ on $s \in [1, 2]$, against the corrected law’s identically zero drift — is the mechanism that separates the two classes (certainty-equivalence and per-cell) on systems where the reading is biased.

Proof. The inclusions follow from monotonicity (each restricted class is contained in the next). The counts and the two separating beliefs are computed exactly by the verification script; the drift identity $g(s + \frac{1}{10}) - g(s) = \frac{s}{5} + \frac{1}{100} \geq \frac{21}{100}$ for $s \geq 1$ is exact — the drift of the *uncorrected* canonical law (laws acting on the biased reading as if it were the state), matching the selector paper’s convention (its corrected law has drift identically zero). $\square$

Proposition 3 (The four-cell grid is a lattice). The two restriction axes — reading (aggregate, full) and action discipline (no-repeat codex, free) — generate four classes whose kernels on the pair family hold exactly 24 (institutional), 26 (aggregate, free), 25 (full, codex), and 28 (full, free) of the 36 safe pairs. The classes carry the product order, and the two middle cells are incomparable: the kernel of the aggregate-free class is not contained in the kernel of the full-codex class (witnesses $\{(1, 2), (3, 1)\}, \{(1, 3), (2, 1)\}$), nor conversely (witness $\{(1, 3), (3, 1)\}$). The kernel meet of the middle cells is the institutional kernel, setwise: $\text{Viab}_{\text{agg}} \cap \text{Viab}_{\text{full, codex}} = \text{Viab}_{\text{inst}}$; their join *within the grid* is the full-information kernel. The four kernels therefore form the Boolean lattice $\{0, 1\}^2$: bottom 24, two incomparable middles 26 and 25, top 28.

Proof. All four counts, the setwise inclusions along both axes, the witnesses, the meet identity (intersection of the computed middle kernels equals the computed institutional kernel, belief by belief), and the join by class order are computed exactly by the verification script (E7–E8). $\square$

Distance to viability

For a nonviable belief and a family of restorative moves, the *distance* is the minimum number of moves that restores viability. On a two-state belief the three minima below are structurally immediate — the audit certifies each exhaustively; the diagnostic content is the move families and the exact counts, not the arithmetic.

Proposition 4 (Distances of the audited belief). For $B^* = \{(1, 2), (2, 1)\}$ under the aggregate reading (non-viable): (i) the minimum sensor refinement restoring viability is exactly one cell split — the unique one-split partition (the two singletons) is viable, and zero splits is the nonviable status quo; (ii) the minimum action-set expansion is exactly one instrument — adding $u = 3$ (protect both patches; a modelled instrument, the minimal formal expansion, not an observed one) restores viability of B^* under the *unchanged* aggregate reading and enlarges the aggregate viable family to all 36 pairs (setwise, strict); (iii) the minimum prior shrinking is exactly one state — either deletion yields a viable belief.

Proof. (i) A two-state belief admits exactly one partition into two nonempty cells; the script verifies the singleton partition viable and the zero-split (merged) verdict non-viable. (ii) With $u = 3$ the per-cell action sets of B^* ’s

single aggregate cell gain a jointly safe instrument; the recursion confirms viability, and the family count $26 \rightarrow 36$ is computed over all pairs. (iii) Both singletons are viable under the unchanged reading; the script verifies both and the nonviability of the whole. $\square$

Verified record (8/8 exact check families)

E1 the setwise chain with counts $24 \subseteq 26 \subseteq 28$; E2 strictness at both links plus the re-verified canonical uncorrected CE drift bound; E3 action-expansion monotonicity with counts $26 < 36$ and a strict instance; E4 the sensor-refinement distance = 1 certified by exhaustion of split counts; E5 the action-expansion distance = 1 with the restored belief; E6 the prior-shrinking distance = 1 with both deletions viable; E7 the four-cell grid of Proposition 3: exact counts 24, 26, 25, 28, setwise inclusions along both axes, and the incomparability witnesses; E8 the kernel meet identity (middle-cell intersection equals the institutional kernel, belief by belief) and the $28 = \binom{8}{2}$ characterization of the top (all pairs avoiding the dynamically dead corner).

What is not claimed

The taxonomy is instantiated on one working system at one belief family (pairs): the lattice is the four audited classes — delayed, history-dependent, and chance-constrained classes remain off-grid — no general computational theory of distances is claimed, no continuous-state version, and no claim that one instrument of protection is a realistic policy lever — it is the minimal formal expansion. Institutional restrictions are modeled only as action codices; procedural and enforcement constraints belong to the programme’s decentralized- observation companion. The stochastic selector’s chance constraints are not revisited. The certainty-equivalence identification is stated for this family only, with the trap cited for the general separation.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `policy_class_lattice_v2_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all eight check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review of the first edition; the joint audit of the first edition and this revision were prepared with further AI assistance, under the programme’s verification discipline.